BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
Work Integrated Learning Programmes Division
First Semester 2024-2025
Mid-Semester Test
(EC-2 Makeup)

Course No.	:	AIMLCZC418
Course Title	:	Introduction to Statistical Methods
Nature of Exam	:	Closed Book
Weightage	:	30%
Duration	:	2 Hours
Date of Exam	:	

No. of Pages = 4
No. of Questions = 6

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Answer all the questions:

Q.1 a) A sample of ten weights (in kg) is given: 30, 32, 35, 38, 40, 42, 45, 48, 50, 53.

i) Calculate Q1, Q2, Q3, and IQR.
ii) Determine potential outliers using the 1.5 × IQR rule. [3 marks]

ANSWER :

Given Data:

Weights: 30,32,35,38,40,42,45,48,50,53 (already sorted).

(i) Calculate Q_1, Q_2, Q_3, and IQR

1. Q_1 (First Quartile):

The first quartile (Q_1) is the median of the lower half of the data (excluding the overall median if the dataset size is odd).

Lower half: $30, 32, 35, 38, 40$.

Median of this subset:

$$Q_1 = 35$$

2. Q_2 (Second Quartile or Median):

The overall median (Q_2) is the middle value of the dataset.

Dataset: $30, 32, 35, 38, 40, 42, 45, 48, 50, 53$.

Since there are 10 values (even number of observations):

$$Q_2 = \frac{40 + 42}{2} = 41$$

3. Q_3 (Third Quartile):

The third quartile (Q_3) is the median of the upper half of the data (excluding the overall median if the dataset size is odd).

Upper half: $42, 45, 48, 50, 53$.

Median of this subset:

$$Q_3 = 48$$

4. Interquartile Range (IQR):

The IQR is the difference between Q_3 and Q_1:

$$IQR = Q_3 - Q_1 = 48 - 35 = 13$$

(ii) Determine potential outliers using the $1.5 \times \mathrm{IQR}$ rule

The $1.5 \times \mathrm{IQR}$ rule states:

- Lower bound: $Q_1 - 1.5 \times \mathrm{IQR}$
- Upper bound: $Q_3 + 1.5 \times \mathrm{IQR}$

1. Calculate bounds:

$$\text{Lower bound} = Q_1 - 1.5 \times \mathrm{IQR} = 35 - 1.5 \times 13 = 35 - 19.5 = 15.5$$

$$\text{Upper bound} = Q_3 + 1.5 \times \mathrm{IQR} = 48 + 1.5 \times 13 = 48 + 19.5 = 67.5$$

2. Identify potential outliers:

Outliers are any data points outside the range $[15.5, 67.5]$.

Given data: $30, 32, 35, 38, 40, 42, 45, 48, 50, 53$.

- No values are below 15.5.
- No values are above 67.5.

Conclusion: There are no potential outliers in this dataset.

Final Answers:

1. **Quartiles and IQR:**
   - $Q_1 = 35$, $Q_2 = 41$, $Q_3 = 48$, $IQR = 13$.
2. **Potential Outliers:** There are no potential outliers in the dataset.

b) Consider the following data from a manufacturing company. Let W be the set of workers in the company. Some subsets of W and their probabilities are provided below. Fill in the missing terms in the table, validate the consistency of the data, and provide justifications. [3 marks]

Set in Words	**Symbol**	**Probability**
All machine operators	M	
All quality inspectors	Q	
All machine operators and/or quality inspectors		0.40
All machine operators who are not quality inspectors		0.15
All quality inspectors who are not machine operators		0.10
All machine operators and quality inspectors		

Solution

Step 1: Relationship Between the Sets
The union of the two sets $M \cup Q$ can be expressed as:
$P(M \cup Q) = P(M - Q) + P(Q - M) + P(M \cap Q)$
Substitute the given values:
$0.40 = 0.15 + 0.10 + P(M \cap Q)$
Solve for $P(M \cap Q)$
$P(M \cap Q) = 0.40 - 0.25 = 0.15$
So, $P(M \cap Q) = 0.15$
Step 2: Calculate P(M)
The set of all machine operators M includes those who are only machine operators $M - Q$ and those who are both machine operators and quality inspectors $M \cap Q$
$P(M) = P(M - Q) + P(M \cap Q)$
Substitute the values:
$P(M) = 0.15 + 0.15 = 0.30$
So, $P(M) = 0.30$
Step 3: Calculate P(Q)
Similarly, the set of all quality inspectors Q includes those who are only quality inspectors ($Q - M$ and those who are both machine operators and quality inspectors ($M \cap Q$
$P(Q) = P(Q - M) + P(M \cap Q)$
Substitute the values:
$P(Q) = 0.10 + 0.15 = 0.25$
So, $P(Q) = 0.25$
Step 4: Validate Consistency
The probabilities must satisfy the following constraint:
$P(M \cup Q) \leq P(M) + P(Q)$
Substitute the values:
$0.40 \leq 0.30 + 0.25$
This is valid. Thus, there are no inconsistencies in the data.
Final Table with Missing Terms

Set in Words	**Symbol**	**Probability**
All machine operators	M	**0.30**
All quality inspectors	Q	**0.25**
All machine operators and/or quality inspectors	$M \cup Q$	0.40
All machine operators who are not quality inspectors	$M - Q$	0.15
All quality inspectors who are not machine operators	$Q - M$	0.10
All machine operators and quality inspectors	$M \cap Q$	**0.15**

Justification
The calculations confirm that all probabilities align with set theory rules and satisfy constraints, ensuring no inconsistencies.

Q2. a) A set of final examination grades in an introductory statistics course is normally distributed, with a mean of 73 and a standard deviation of 8.

i. What is the probability of getting a grade below 91 on this exam?
ii. What is the probability that a student scored between 65 and 89? [3 Marks]

ANSWER :

To solve these problems, we use the properties of the normal distribution. The given data is:

- Mean (μ) = 73
- Standard Deviation (σ) = 8

The standard normal variable (Z) is calculated as:

$$Z = \frac{X - \mu}{\sigma}$$

Where X is the value of interest.

(i) Probability of getting a grade below 91

We calculate Z for $X = 91$:

$$Z = \frac{91 - 73}{8} = \frac{18}{8} = 2.25$$

Using standard normal distribution tables or a calculator, the cumulative probability for $Z = 2.25$ is approximately:

$$P(Z < 2.25) = 0.9878$$

Thus, the probability of getting a grade below 91 is:

P(X < 91) = 0.9878

(ii) Probability that a student scored between 65 and 89

We calculate Z for both $X = 65$ and $X = 89$:

For $X = 65$:

$$Z = \frac{65 - 73}{8} = \frac{-8}{8} = -1$$

For $X = 89$:

$$Z = \frac{89 - 73}{8} = \frac{16}{8} = 2$$

Using standard normal distribution tables or a calculator:

- $P(Z < -1) = 0.1587$
- $P(Z < 2) = 0.9772$

The probability of scoring between 65 and 89 is:

$$P(65 \leq X \leq 89) = P(Z < 2) - P(Z < -1)$$

= 0.9772 - 0.1587 = 0.8185

= 81.85%

Final Answers:

1. **Probability of getting a grade below 91:** 0.9878 (98.78%).
2. **Probability of scoring between 65 and 89:** 0.81850.

b) One-fourth of the residents of the Ram Nagar Estates leave their garage doors open when they are away from home. The local chief of police estimates that 5 percent of the garages with open doors will have something stolen, but only 1 percent of those closed will have something stolen. If a garage is robbed, what is the probability the doors were left open. [3 Marks]
ANSWER:

To solve this problem, we use **Bayes' Theorem:**

$$P(\text{Open} \mid \text{Robbed}) = \frac{P(\text{Robbed} \mid \text{Open}) \cdot P(\text{Open})}{P(\text{Robbed})}$$

Definitions:

1. $P(\text{Open})$: Probability the garage door is left open = 0.25.
2. $P(\text{Closed})$: Probability the garage door is closed = $1 - 0.25 = 0.75$.
3. $P(\text{Robbed} \mid \text{Open})$: Probability a garage is robbed given its door is open = 0.05.
4. $P(\text{Robbed} \mid \text{Closed})$: Probability a garage is robbed given its door is closed = 0.01.

Total Probability of a Robbery ($P(\text{Robbed})$):

$$P(\text{Robbed}) = P(\text{Robbed} \mid \text{Open}) \cdot P(\text{Open}) + P(\text{Robbed} \mid \text{Closed}) \cdot P(\text{Closed})$$

Substitute Values:

$$P(\text{Robbed}) = (0.05 \cdot 0.25) + (0.01 \cdot 0.75)$$

Probability the door was left open ($P(\text{Open} \mid \text{Robbed})$):

$$P(\text{Open} \mid \text{Robbed}) = \frac{P(\text{Robbed} \mid \text{Open}) \cdot P(\text{Open})}{P(\text{Robbed})}$$

1. **Total Probability of a Robbery ($P(\text{Robbed})$):** 0.02 or 2%.
2. **Probability the door was left open given a robbery occurred ($P(\text{Open} \mid \text{Robbed})$):** 0.625 or 62.5%.

3 Q a) A retail bank classifies its customers based on their income levels into three categories: **Low Income**, **Middle Income**, and **High Income**. The table below shows the number of customers who have defaulted or not defaulted on their loans:

Loan Default	**Low Income**	**Middle Income**	**High Income**	**Total**
No	8000	15000	7000	30000
Yes	2000	5000	1000	8000
Total	10000	20000	8000	38000

i) What is the probability that a person will **not default on the loan** given they belong to the **Middle Income** group?

ii) What is the probability that a person belongs to the **Middle Income** group given they have not defaulted on the loan? [3 Marks]

Given Data:

Loan Default	Low Income	Middle Income	High Income	Total
No	8000	15000	7000	30000
Yes	2000	5000	1000	8000
Total	10000	20000	8000	38000

Question 1: $P(\text{No Default} \mid \text{Middle Income})$

Using the conditional probability formula:

$$P(\text{No Default} \mid \text{Middle Income}) = \frac{P(\text{No Default and Middle Income})}{P(\text{Middle Income})}$$

From the table:

- $P(\text{No Default and Middle Income}) = 15000$
- $P(\text{Middle Income}) = 20000$

$$P(\text{No Default} \mid \text{Middle Income}) = \frac{15000}{20000} = 0.75$$

So, the probability is **0.75 or 75%**.

Question 2: $P(\text{Middle Income} \mid \text{No Default})$

Using the conditional probability formula:

$$P(\text{Middle Income} \mid \text{No Default}) = \frac{P(\text{No Default and Middle Income})}{P(\text{No Default})}$$

From the table:

- $P(\text{No Default and Middle Income}) = 15000$
- $P(\text{No Default}) = 30000$

$$P(\text{Middle Income} \mid \text{No Default}) = \frac{15000}{30000} = 0.5$$

So, the probability is **0.5 or 50%**.

Final Answers:

1. $P(\text{No Default} \mid \text{Middle Income}) = 75\%$
2. $P(\text{Middle Income} \mid \text{No Default}) = 50\%$

b) A biased die has a probability of rolling a "6" equal to p=0.2. The die is rolled n=25 times. Calculate the probability of getting exactly k=3 of "6" using both the Binomial and Poisson assumptions. Compare the results. [3 Marks]

ANSWER

Given:

- Probability of rolling a 6 (p) = 0.20
- Number of rolls (n) = 25,
- Desired number of 6s (k) = 3

We will calculate P(X=3) using both the **Binomial** and **Poisson** assumptions.

1. Binomial Distribution

The probability mass function (PMF) of a Binomial distribution is:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Where:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Substitute the values $n = 25$, $k = 3$, $p = 0.2$:

$$P(X = 3) = \binom{25}{3} (0.2)^3 (1 - 0.2)^{25-3}$$

First, calculate:

- $\binom{25}{3} = \frac{25 \times 24 \times 23}{3 \times 2 \times 1} = 2300,$
- $(0.2)^3 = 0.008,$
- $(0.8)^{22} \approx 0.0724$ (calculated using a calculator).

$$P(X = 3) = 2300 \times 0.008 \times 0.0724 = 0.1334$$

So, using the **Binomial distribution:**

$$P(X = 3) = 0.1334 \text{ (or } 13.34\%).$$

2. Poisson Approximation

The Poisson distribution is an approximation to the Binomial distribution when n is large, p is small, and $np = \lambda$. Here:

$$\lambda = n \cdot p = 25 \cdot 0.2 = 5$$

The PMF of the Poisson distribution is:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Substitute $\lambda = 5$, $k = 3$:

$$P(X = 3) = \frac{5^3 e^{-5}}{3!}$$

First, calculate:

- $5^3 = 125$,
- $e^{-5} \approx 0.0067$,
- $3! = 6$.

$$P(X = 3) = \frac{125 \cdot 0.0067}{6} = \frac{0.8375}{6} \approx 0.1396$$

So, using the **Poisson approximation**:

$$P(X = 3) = 0.1396 \text{ (or } 13.96\%\text{)}.$$

Comparison

- **Binomial result:** $P(X = 3) = 0.1334$ (13.34%).
- **Poisson result:** $P(X = 3) = 0.1396$ (13.96%).

The Poisson approximation is close to the Binomial result but slightly overestimates the probability due to the approximation assuming n is very large. Here, $n = 25$ is moderately large, so the approximation is reasonable.

Final Answer

1. **Binomial**: $P(X = 3) = 0.1334$.
2. **Poisson**: $P(X = 3) = 0.1396$.
3. The Poisson approximation slightly overestimates the probability compared to the Binomial distribution.

Q 4 a) The following table interprets the promotion type and the corresponding customer purchases. If the promotion type is **Social Media**, what is the probability that the customer will purchase the product? Use Naïve Bayes Theorem.

Promotion Type	Customer Purchases	Promotion Type	Customer Purchases
Email	Yes	Email	Yes
Social Media	No	TV Ad	Yes
TV Ad	Yes	Social Media	No
Email	No	TV Ad	Yes
Social Media	Yes	Email	No
Email	Yes	Social Media	Yes
TV Ad	No	Email	Yes
Social Media	No	------	---

[3 Marks]

ANSWER:

To determine the likelihood that a customer will purchase a product when the promotion type is "Social Media," we need to analyze the given data. Here's the solution step by step:

Step 1: Extract data related to "Social Media"

From the table provided:

1. **Social Media → No**
2. **Social Media → No**
3. **Social Media → Yes**
4. **Social Media → Yes**
5. **Social Media → No**

So, for "Social Media," we have:

- **Yes**: 2 occurrences
- **No**: 3 occurrences

Step 2: Calculate the likelihood

The likelihood (probability) of a customer purchasing the product given "Social Media" promotion is:

$$\text{Likelihood} = \frac{\text{Number of "Yes"}}{\text{Total occurrences of Social Media}}$$

$$\text{Likelihood} = \frac{2}{2+3} = \frac{2}{5}$$

The likelihood that a customer will purchase the product when the promotion type is "Social Media" is **40%**.

b) A data scientist is working on a machine learning model that predicts house prices based on various features such as size, location, and number of bedrooms. The model has some error in its predictions, represented by the random variable X. The error X follows the following probability density function (PDF):

$$f(x) = \begin{cases} \frac{k}{6} & -3 \le x \le 3 \\ 0 & otherwise \end{cases}$$

Calculate $i)$ the value of k , $ii)$ the expected value of error $iii)$ Compute the variance of the error. [3 Marks]

ANSWER:

i) For a valid probability distribution, the total probability over the range must equal 1.

$$\therefore \int_{-3}^{3} f(x)dx = 1 \rightarrow \int_{-3}^{3} \frac{k}{6} dx = 1 \rightarrow \frac{k}{6}\big(3-(-3)\big) = 1 \rightarrow k = 1.$$

$$ii)\ E(X) = \int_{-3}^{3} x.\frac{1}{6}\,dx = \left[\frac{x^2}{12}\right]_{-3}^{3} = 0$$

$$iii)\ Var(x) = E(X^2) - \big(E(X)\big)^2.$$

$$E(X^2) = \int_{-3}^{3} \frac{x^2}{6}\,dx = \left[\frac{x^3}{18}\right]_{-3}^{3} = \frac{1}{6}\left(\frac{27}{3} - \left(-\frac{27}{3}\right)\right) = \frac{18}{6} = 3\,.$$

$$Var(X) = 3 - (0)^2 = 3.$$

Q5 If the joint probability mass function of two discrete random variables is given below:

X	Y			
	0	1	2	3
0	0.11	0.09	0.01	0.07
1	0.09	0.11	0.06	0.01
2	0.01	0.06	0.11	0.04
3	0.07	0.01	0.04	0.11

(i) Find the marginal probability mass function of X and Y
(ii) Find the conditional probability of X = 1 given Y = 2
(iii) Find the conditional probability of Y = 1 given X = 1

(iv) Find the Mean and Variance for the marginal distribution of X and Y [6 Marks]

Given: Joint probability mass function (PMF)

$X\backslash Y$	0	1	2	3	Row Sum ($P_X(X=x)$)
0	0.11	0.09	0.01	0.07	0.28
1	0.09	0.11	0.06	0.01	0.27
2	0.01	0.06	0.11	0.04	0.22
3	0.07	0.01	0.04	0.11	0.23
Column Sum ($P_Y(Y=y)$)	0.28	0.27	0.22	0.23	1.0

(i) Marginal PMFs of X and Y

The marginal PMFs are calculated by summing the probabilities along rows ($P_X(X=x)$) and columns ($P_Y(Y=y)$).

- **Marginal PMF of X:**

$$P_X(X=x) = \text{Row Sum for each } x.$$

$X=x$	$P_X(X=x)$
0	0.28
1	0.27
2	0.22
3	0.23

- **Marginal PMF of Y:**

$$P_Y(Y=y) = \text{Column Sum for each } y.$$

$Y=y$	$P_Y(Y=y)$
0	0.28
1	0.27
2	0.22
3	0.23

(ii) Conditional Probability $P(X = 1 \mid Y = 2)$

The conditional probability is given by:

$$P(X = 1 \mid Y = 2) = \frac{P(X = 1, Y = 2)}{P_Y(Y = 2)}.$$

From the table:

- $P(X = 1, Y = 2) = 0.06,$
- $P_Y(Y = 2) = 0.22.$

$$P(X = 1 \mid Y = 2) = \frac{0.06}{0.22} = 0.2727 \text{ (or } 27.27\%\text{)}.$$

(iii) Conditional Probability $P(Y = 1 \mid X = 1)$

The conditional probability is given by:

$$P(Y = 1 \mid X = 1) = \frac{P(X = 1, Y = 1)}{P_X(X = 1)}.$$

From the table:

- $P(X = 1, Y = 1) = 0.11,$
- $P_X(X = 1) = 0.27.$

$$P(Y = 1 \mid X = 1) = \frac{0.11}{0.27} = 0.4074 \text{ (or } 40.74\%\text{)}.$$

(iv) Expected Value $E(X)$ and Variance $V(X)$

- Expected Value $E(X)$:

$$E(X) = \sum_x x \cdot P_X(X = x).$$

Using the marginal PMF of X:

$$E(X) = (0 \cdot 0.28) + (1 \cdot 0.27) + (2 \cdot 0.22) + (3 \cdot 0.23).$$

$$E(X) = 0 + 0.27 + 0.44 + 0.69 = 1.4.$$

- **Variance $V(X)$:**

$$V(X) = E(X^2) - [E(X)]^2.$$

First, calculate $E(X^2)$:

$$E(X^2) = \sum_x x^2 \cdot P_X(X = x).$$

$$E(X^2) = (0^2 \cdot 0.28) + (1^2 \cdot 0.27) + (2^2 \cdot 0.22) + (3^2 \cdot 0.23).$$

$$E(X^2) = 0 + 0.27 + 0.88 + 2.07 = 3.22.$$

Now, calculate $V(X)$:

$$V(X) = E(X^2) - [E(X)]^2 = 3.22 - (1.4)^2 = 3.22 - 1.96 = 1.26.$$

Final Answers:

1. **Marginal PMFs:**
   - $P_X(X = x)$: $0.28, 0.27, 0.22, 0.23.$
   - $P_Y(Y = y)$: $0.28, 0.27, 0.22, 0.23.$
2. $P(X = 1 \mid Y = 2) = 0.2727$ (27.27%).
3. $P(Y = 1 \mid X = 1) = 0.4074$ (40.74%).
4. $E(X) = 1.4,\ V(X) = 1.26.$